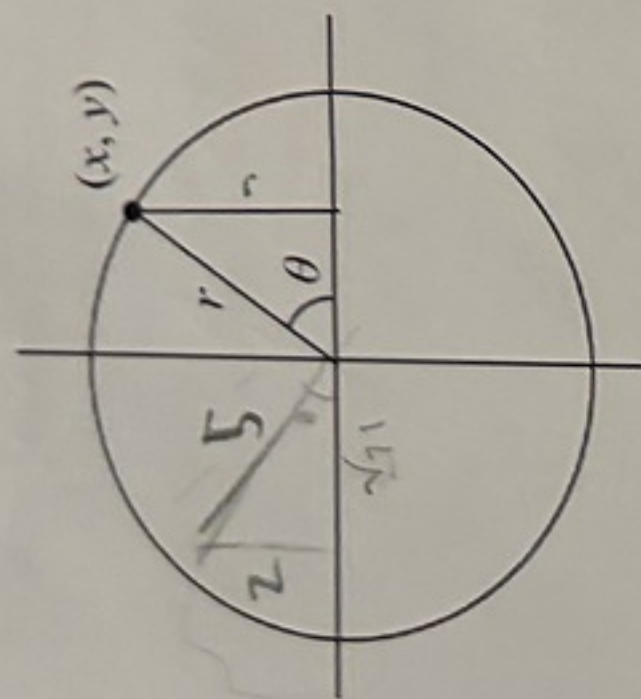


Derive and apply the Law of Cosines.

Derive and apply the Law of Sines and Law of Extended Sines

1. Here is a circle centered at the origin with radius r . The point on the circle indicated by the angle θ has coordinates (x, y) . Find the coordinates of the point in terms of trig functions. Explain!



$$\sin(\theta) = \frac{y}{r}$$

$$\cos(\theta) = \frac{x}{r}$$

$$r = \sec(\theta)$$

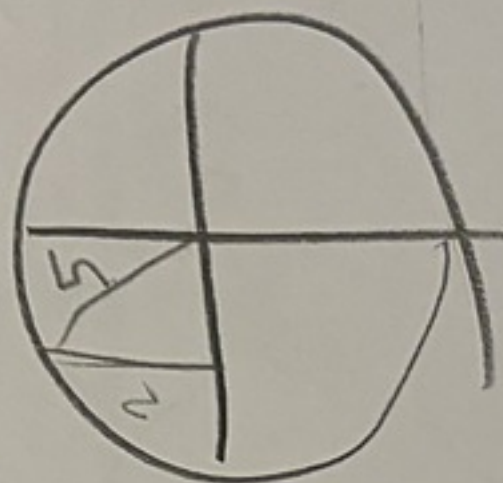
2. If $\sin(\theta) = \frac{2}{5}$, and θ is in quadrant 2, then find:

a. $\cos(\theta) =$

$$\frac{-\sqrt{21}}{5}$$

b. $\tan(\theta) =$

$$\frac{2}{-\sqrt{21}} = \frac{2}{-5\sqrt{21}}$$



UNIT 8: Trigonometry

Handbook: Law of Sines, Law of Cosines, Law of Tangents



7. If $\sin(\theta) = \frac{3}{5}$ and $\tan(\theta) < 0$, then

a. What quadrant is the angle θ in?

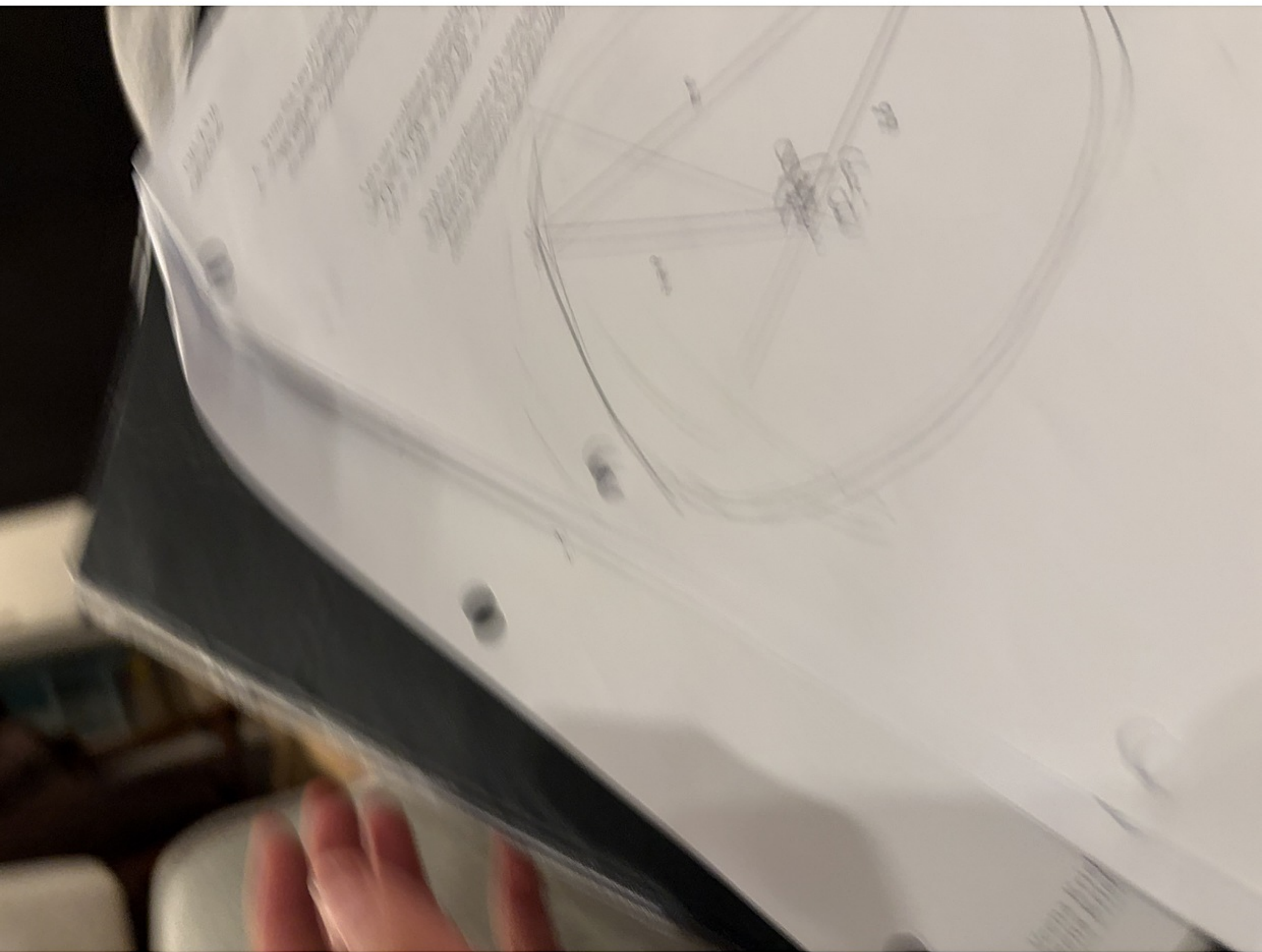
and

b. Calculate $\cos(\theta)$

$$\frac{\sqrt{5}}{4}$$

c. Calculate $\tan(\theta)$

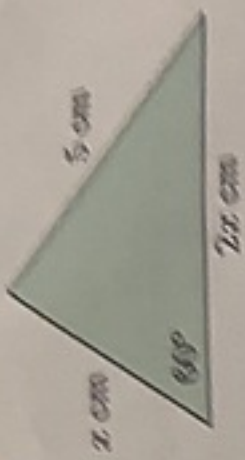
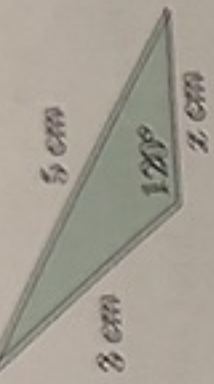
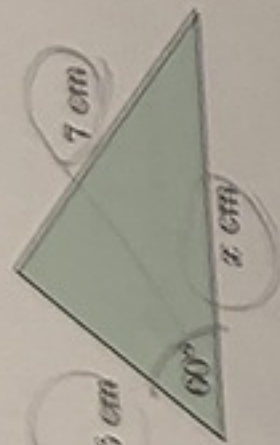
$$\frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$$



Handout 8: Law of Cosines, Law of Sines, Law of Tangents

$$c^2 = a^2 + b^2 - 2ab \cos(C)$$

5. Solve for the value of x in each of the following triangles.



$$49 = 36 + x^2 - 3x$$

$$x^2 - 3x - 13 = 0$$

$$x = \frac{3 \pm \sqrt{9 + 52}}{2}$$

$$x = \frac{3 + \sqrt{61}}{2}$$

$$x = \frac{-3 + \sqrt{9 + 52}}{2}$$

$$25 = 9 + x^2 - 3x$$

$$3 \pm \sqrt{9 + 16} = 3 \pm 5$$

$$2 \pm \sqrt{17} = 0$$

$$x = \frac{-3 \pm \sqrt{9 + 16}}{2}$$

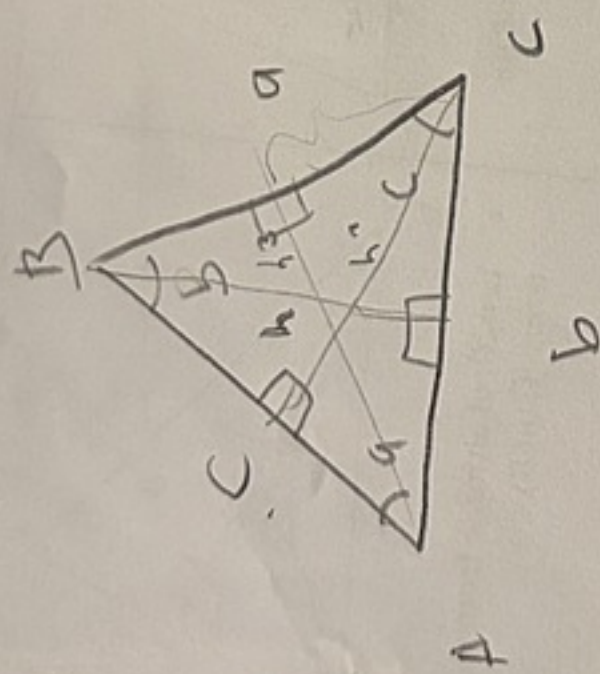
$$25 = x^2 + 4x^2 + 2x^2$$

$$25 = 6x^2$$

$$x = \frac{\sqrt{50}}{6}$$

6. Show that for any standard triangle $\triangle ABC$, that

$$\frac{\cos(A)}{a} + \frac{\cos(B)}{b} + \frac{\cos(C)}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$



$$a^2 = a^2 + b^2 - 2ab \cos(C) + a^2 + c^2 - 2ac \cos(B)$$

$$2cb \cos(A)$$

$$- 2ab \cos(C) + 2ac \cos(B) + 2cb \cos(A) =$$

$$a^2 + b^2 + c^2$$

Use cosines to solve for each

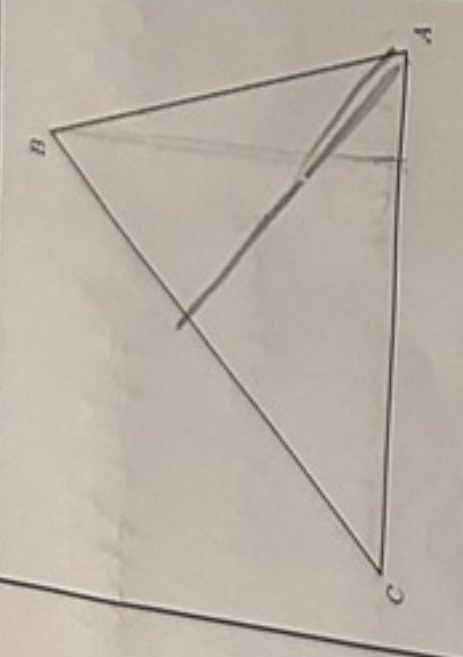
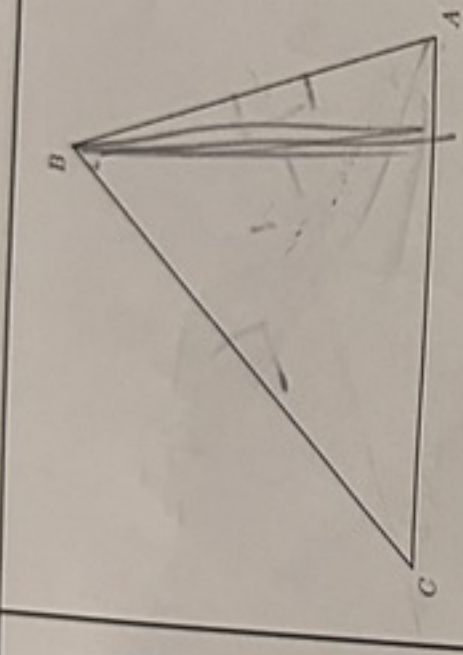
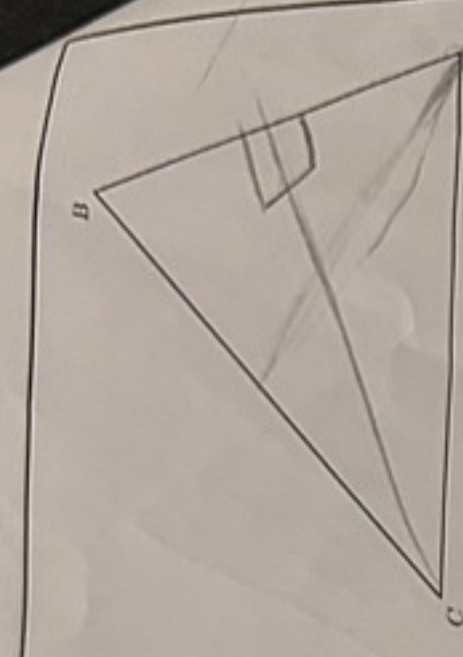
$$\frac{a^2}{c} = \frac{c^2 + b^2 - 2cb \cos(A)}{c}$$

$$a^2 = \frac{c^2 + b^2 - 2cb \cos(A)}{c}$$

$$b^2 = \frac{a^2 + c^2 - 2ac \cos(B)}{b}$$

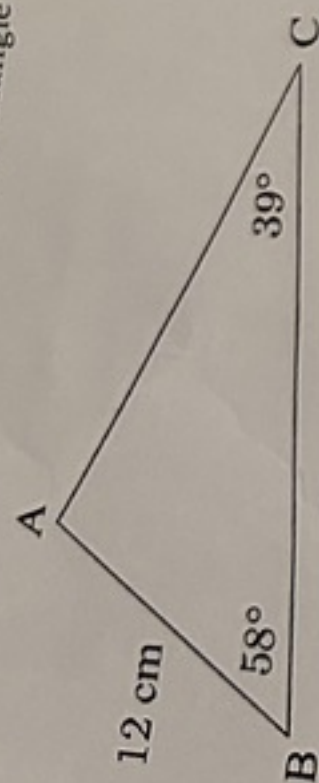
$$\frac{a^2 + b^2 + c^2}{2abc} = \frac{\cos(C)}{c} + \frac{\cos(B)}{b} + \frac{\cos(A)}{a}$$

7. Given triangle $\triangle ABC$, such that $CA = b$, $CB = a$, and $AB = c$, Law of Sines states that $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$. Draw the altitudes of the triangle from each vertex, and find their lengths using right triangle trigonometry:

		
Length of altitude from A (in terms of $\sin(C)$) $h_a = b \sin(C)$	Length of altitude from B (in terms of $\sin(A)$) $h_b = a \sin(A)$	Length of altitude from C (in terms of $\sin(B)$) $h_c = a \sin(B)$

- Calculate the area of the triangle in three different ways, and set your three area expressions equal to each other to find three equations (usually rolled into one, like $x = y = z$). Then show that expression is the same as the one above (the statement of the Law of Sines)

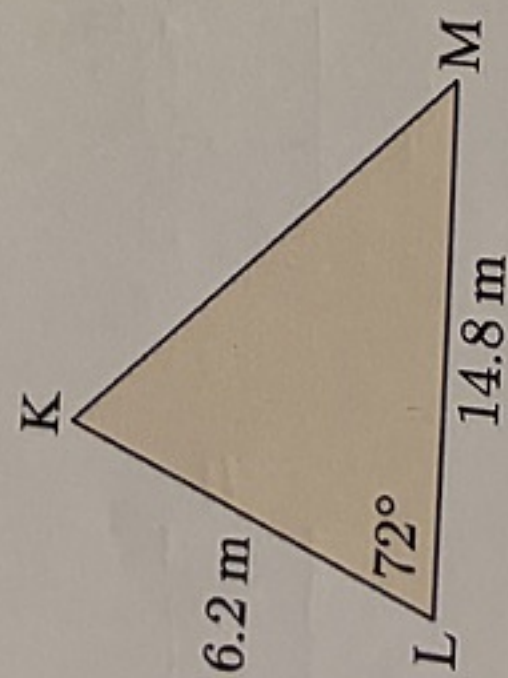
8. Find the length of side $\overline{AC}$ in the triangle below.



$$\frac{12}{\sin(39)} = \frac{AC}{\sin(58)}$$

$$AC = \frac{12 \sin(58)}{\sin(39)} = 16.17$$

9. Find the length KM



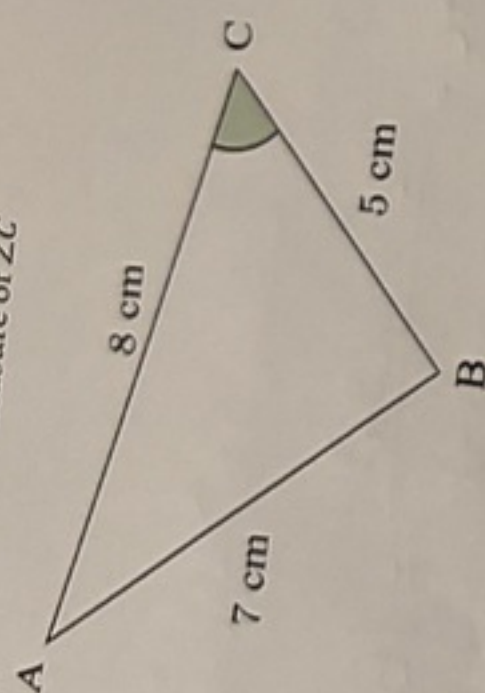
$$c^2 = a^2 + b^2 - 2ab \cos(72)$$

$$\frac{KM}{\sin(K)} =$$

$$KM = (6.2)^2 + (14.8)^2 - 2 \cdot 6.2 \cdot 14.8 \cos(72)$$

$$KM = 14.16$$

10. Find the measure of $\angle C$



$$49 = 64 + 25 - 40 \cos(C)$$

$$30 = 40 \cos(C)$$

$$\frac{3}{4} = \cos(C)$$

$$\cos^{-1}\left(\frac{3}{4}\right) = \angle C = 41.4^\circ$$